



#	Milestone	Generative AI Guidelines	Due	Mark Release
1	Project Definition & Team Formation	C: AI-Assisted Editing ¹	Sep. 23, Wed., AoE ²	Sep. 28, Mon.

This course is *project driven*. Students will work in teams to define, develop, and complete a project in the field of data mining/machine learning over the course of one semester. Each team will consist of undergraduate students and one graduate student, who serves as the project lead, coordinating the project and guide the team through the research and development process. The course project has two main objectives:

- Hands-on experience: Gain practical experience in applying data mining and machine learning methods to a real-world problem, while developing skills in project planning, implementation, evaluation, and teamwork.
- Research experience: Develop an understanding of the research *process*, from identifying a problem and reviewing existing approaches to designing, implementing, and evaluating a solution.

Projects should aim to advance the state of the art, rather than simply reproduce an existing solution. This provides students with an opportunity to conduct meaningful research while completing the course requirements.

A research project typically starts with a problem and proposes a solution by designing, implementing, and evaluating an algorithm or computational approach. In this sense, it may resemble a software project; however, there are important differences. A software project may primarily focus on implementing or integrating existing algorithms to build a functional system. In contrast, a research project should aim to investigate a problem, identify a limitation or gap in existing approaches, and propose a solution that addresses that gap. The proposed solution may involve (1) a new algorithm, (2) a modification or extension of an existing algorithm, (3) a new combination/pipeline of existing techniques, or (4) a novel application of existing methods to a problem where they have not been adequately explored. Therefore, the goal is not simply to implement an existing method, but to make a meaningful contribution beyond what has already been done, supported by appropriate experiments and evaluation. Roughly, a research project consists of the following milestones (phases):

1. Project Definition & Team Formation (Proposal)
2. Background & Related Work (Literature Review)
3. Solution Design & Implementation (Proposed Method)
4. Evaluation (Experimentation)
5. Project Presentation: 3-Minute Talk (3MT)
6. Project Management & Documentation (Graduate Student Only, Umbrella Activity)

A project manual has been prepared to guide students through each milestone. Each milestone breaks the project into a series of smaller steps and deliverables. The current manual covers Milestone 1: Project Definition & Team Formation, also known as Proposal.

1.1. Define a Data Mining/Machine Learning Problem

The 1st step of the Proposal is to select and define a problem in data mining/machine learning. In many cases, a problem has already been identified, and different solutions have been proposed to address it. This does not mean that the problem is solved. Established problems remain open to new or improved solutions that can achieve

¹ AI can be used to make improvements to the clarity or quality of student-created work to improve the final output, but no new content can be created using AI (e.g. language editing for lab reports). AI can be used, but your original work with no AI content must be provided in an appendix.

² <https://www.timeanddate.com/time/zones/aoe>

better results, reduce computational cost, improve scalability or robustness, simplify implementation, or address limitations that existing methods overlook. For example, consider “*sentiment analysis of tweets*”. A survey like the one by Giachanou and Crestani³ reviews a range of approaches that have been proposed for this problem nearly *ten* years ago. A project does not necessarily need to identify an entirely new problem. Instead, it may investigate whether existing approaches can be improved or adapted to address a particular limitation or setting.

You may initially come up with a problem that appears to be completely original. However, after conducting a more thorough search, you will often discover that a similar problem has already been studied, perhaps under a different name or in a different context. This is not a reason to abandon the idea. Finding related work is an important part of project and can help you identify how your problem differs from what has already been studied.

For instance, suppose you propose “*detecting the author's gender from tweets*.” At first, you may think that this is a new problem. A quick search, however, reveals existing work on “*author-gender detection as a document-classification problem*.” This does not necessarily mean that your project is no longer novel. The existing methods may have been developed for a substantially different type of document, such as books. Tweets are typically much shorter, informal, and context-dependent than books. Therefore, you could investigate whether methods designed for longer and more formal documents perform poorly on tweets and propose an approach that addresses the specific characteristics of short, informal social-media text.

In other words, novelty does not necessarily mean inventing a completely new problem. A contribution may instead come from applying an existing solution to a new setting, identifying a limitation, modifying an existing method, combining existing techniques in a new way, or developing a method that performs better under constraints. At the same time, there are problems for which *no* solution has yet been proposed, or for which existing solutions have consistently been unsuccessful. Although such problems can be highly interesting, they are generally difficult to evaluate in a course project. *Therefore, you should not select such a problem for this course without first obtaining the lab instructor's approval.*

1.2. Establish Motivation and Significance

The 2nd step of the Proposal is to explain why the chosen problem is worth solving. We should not only ask “*Can we solve this problem?*” but also “*Why does solving this problem matter?*”

You should identify a practical, scientific, or societal motivation for your project. In other words, ask:

- *So what, who cares?*
- *Why should we and others put effort into solving this problem?*
- *Who could benefit from a solution?*
- *What could become possible if the problem were solved or existing solutions were improved?*

For example, suppose the problem is “*detecting the gender of the author of a tweet*.” You should explain why this capability could be useful. For example,

“Existing research may show that author-gender information can potentially support

- *Improved product recommendation [citations];*
- *Better friend or user recommendation [citations]; and*
- *Health-related prediction or analysis [citations].”*

Important: Claims about potential benefits should be supported by appropriate scientific references/citations.

³ Anastasia Giachanou and Fabio Crestani. 2016. Like It or Not: A Survey of Twitter Sentiment Analysis Methods. ACM Comput. Surv. 49, 2, Article 28 (June 2017), 41 pages. <https://doi.org/10.1145/2938640>

1.2.1. Motivating Example: At least one concrete example that demonstrates the potential benefit of solving the problem to make the motivation easy to understand by showing a problem situation, the proposed capability, and the potential benefit. For example:

“Consider the tweet “Don’t know what rain jacket to buy,” posted by @princess. If a recommendation system could infer relevant demographic information about the author, it could potentially use that information to provide more personalized recommendations.”

1.2.2. Gaps in Existing Solutions: Finally, you must explain why existing solutions are not sufficient and, consequently, why your project is worth pursuing. There are several possibilities:

- Existing methods address a related problem but not your particular setting;
- Existing methods make assumptions that do not hold in your setting;
- Existing methods perform poorly under particular conditions;
- Existing methods are computationally expensive or difficult to deploy;
- Existing methods overlook an important characteristic of the data;

For example, after reviewing the literature, you might write:

“Although methods have been proposed for identifying an author’s gender from books [citation] and scholarly publications [citation], tweets have substantially different characteristics. Tweets are typically short, noisy, informal, and context-dependent. Therefore, methods designed for longer and more formal documents may not perform equally well on tweets.”

Alternatively, you may discover that methods already exist for tweets but have an important limitation:

“Although several methods have been proposed for identifying an author’s gender from tweets [citation], method A relies on the author’s name as an important source of gender information. This approach may be unreliable when the author’s name is gender-neutral (e.g., “Andy”) or when the username provides no meaningful gender information (e.g., “Empr_World”). This limitation motivates an approach that does not depend on the author’s name.”

In summary, a strong proposal should provide convincing answers to two distinct questions for its motivation:

- Why is the problem important? → What is the potential benefit of solving it?
- Why is further research necessary? → What limitation, gap, or opportunity remains in existing solutions?

1.3. Problem Definition

The 3rd step of the Proposal is to formally define the problem in clear, precise, and unambiguous language. The definition should be understandable to people in the field and should specify exactly what is given as input, what needs to be produced as output, and what the task is required to accomplish. Next to the intuitive explanation of the problem, you should use mathematical notation to formalize the problem. For example, consider the problem of identifying the gender of the author of a tweet. An intuitive description might be:

“Given a tweet, determine whether its author is male or female.”

A more formal definition would be:

“Author Gender Identification per Tweet: Given a corpus M and a tweet $t \in M$ posted by a user u , the task is to learn a mapping function $g(t): M \rightarrow \{0, 1\}$ from a set of tweets to the Boolean set such that $g(t)=0$ if u is male and 1 otherwise.”

*Do not mix the problem definition with the solution. The problem definition specifies what needs to be solved, not how it should be solved. Do not describe a particular algorithm, model, architecture, feature set, or implementation technique. These belong to **Milestone 3: Solution Design & Implementation**. Exceptionally, for this course, formulate the problem as “learning a function”, since the proposed solutions must use data mining/machine learning methods. But whether the function (e.g., $g(t)$) is learned using an svm, decision tree, neural network, graph neural network, transformer, or another data mining/machine learning techniques should not be discussed here.*

1.3.1. Runing Example: Your formal problem definition should be followed by an intuitive example that helps the reader understand what the formal definition means in practice. For example:

“Given the tweet t : “Hey Georgia! If you still have your mail-in ballot, return it to a drop-off location before 7 PM ET today. Find one near you at <http://weall.vote/ga>. Let’s get it done! Police cars revolving light,” posted by @MichelleObama, the task is to predict the author’s gender, represented as $g(t)=1$ (female).”

The example should demonstrate the relationship between the input and the expected output without introducing additional assumptions that are not part of the formal problem definition. Your formal problem definition should be precise enough that the task could be implemented solely from its formal description.

1.4. Author List

The 4th purpose of the Proposal is to create a team of a minimum of 2 and maximum of 3 undergrad students, and 1 grad student. Individual projects are highly discouraged and are permitted only in rare and exceptional cases. Team projects facilitate better workload distribution, earlier and more effective fault detection, and opportunities to develop collaboration and conflict-resolution skills. In a team project:

- The rationale for the members’ collaboration must be justified, such as through previous successful collaboration, shared experience in the research topic, or complementary expertise; and
- Each member’s contribution to every part of the research project must be clearly identified. Evaluation scores will be distributed among team members based on their individual contributions.

1.5. Submission Guidelines

1.5.1. Deliverables

- Submission must be written in English, in the current ACM two-column conference format in LaTeX. Overleaf⁴ templates are available from the ACM Website⁵. Use the *sigconf* proceedings template.
- The submission must be *exactly* 1 page in length, no more and no less, including author list, figures, tables, and references, with unlimited pages for C: AI-Assisted Editing appendix, and must be authored by all team members, with the graduate student listed last in the author list. We understand that fitting all

⁴ <https://www.overleaf.com/gallery/tagged/acm-official>

⁵ <https://www.acm.org/publications/proceedings-template>

Commented [A1]: Hint: The variable (u) has never been used and can therefore be safely removed. But see my next comments below.

Commented [A2]: Hint: This problem only looks at raw text, overlooking other information available about the user, such as their social network. For example, if most of a user’s friends are female, this information may provide additional evidence suggesting that the user is more likely to be female. Such additional sources of information can be used to complement the information contained in the raw text.

Commented [A3]: Hint: Mathematically, the variable (u) can be replaced with (t), but it might be confusing for the general reader as text does not have gender.

the required content into exactly one page can be challenging. However, this is intentional: an important part of communication is learning how to retouch, refine, condense, and prioritize information without losing its essential meaning. We recommend initially writing a longer version (e.g., 3-5 pages) and then gradually reducing and retouching it until you can effectively communicate the key ideas within the required one-page limit.

- The submission must be provided as a single zip file named: 1_Proposal_firstname1_firstname2_[...].zip, e.g., 1_Proposal_Reia_Ali_Luiz_Hossein.zip, containing:
 - o The LaTeX source files and any required supplementary files;
 - o The compiled one-page PDF file.

A *sample* is attached to this manual and is also available [here](#)⁶ (01_Proposal.tex).

1.5.2. Team and Individual Contributions

- Each team must maintain a GitHub repository for the project and linked to the Abstract. The repository must be associated with the individual GitHub accounts of all team members. The repository will be used to document the development of the project and to make individual contributions traceable. The GitHub repository should be used throughout the project rather than only immediately before a submission deadline. The lab instructor and/or marker may consider the repository's commit history, pull requests, issues, documentation, and other project records when assess individual contributions. The number of commits alone will not be used as a measure of contribution. The quality, significance, and nature of each member's contributions will also be considered.
- The graduate student serves as the project lead, responsible for coordinating the process, maintaining the overall project direction, and ensuring that the project develops toward a meaningful contribution. However, the project lead is also expected to be an active contributor, participating in activities such as problem formulation, literature review, methodology development, implementation, experimentation, analysis, and writing.
- Undergraduate members contribute to the project activities under the guidance of the project lead. All team members are expected to make substantive contributions, and no member should be limited solely to coordination or project management.
- As part of **Milestone 6. Project Management & Documentation (Graduate Student Only)**, the graduate student (project lead) is expected to document each member's contributions throughout the project, with a brief contribution statement provided at each major project milestone. These statements will be compiled into the final project management report.

1.5.3. Research Problem Selection

You may not select the same research problem as another group unless you are members of the same team. Once you have identified a potential research problem, communicate your choice to the lab instructor and obtain approval before proceeding to the next step.

1.5.4. Marking Scheme

- 10% Abstract
- 10% Introduction
- 20% Motivation for solving the problem
- 10% Motivating example
- 20% Formal problem definition
- 10% An example instance of formal problem
- 10% Team members (author list, including affiliation) + collaboration justification
- 10% Style, format, GitHub repo, file naming convention, ...

⁶ <https://www.overleaf.com/read/bsrxtwqgggsb>